ChemCritic 文献证据卡

PMC6941188 | 氧化 / 环氧化 | 采集状态 imported_unchecked

标题	Visible Light-Emitting Diode Light-Driven Cu0.9Fe0.1@RCAC-Catalyzed Highly Selective Aerobic Oxidation of Alcohols and Oxidative Azo-Coupling of Anilines: Tandem One Pot Oxidation – Condensation to Imidazoles and Imines
作者	Patel A., Patel G., Banerjee S.
出处	ACS Omega 2019
DOI	10.1021/acsomega.9b03096
原文链接	https://pmc.ncbi.nlm.nih.gov/articles/PMC6941188/
许可	This is an open access article published under an ACS AuthorChoice License, which permits copying and redistribution of

摘要（原文节选）

Here, we have demonstrated visible light-emitting diode light-driven selective and efficient aerobic oxidation of primary/secondary alcohols to aldehydes/ketones and oxidative azo-coupling of anilines using biomass rice husk-derived chemically activated carbon sheet-supported copper – iron bimetallic hybrid nanomaterials (CuxFe1 – x@RCAC) under oxidant and additive-free conditions. The catalytic activity of the CuxFe1 – x@RCAC materials has been investigated for the oxidation of alcohols and anilines, and Cu0.9Fe0.1@RCAC ⋯

数据表 · Table 2（15 行，含数值行 15 行）

Optimization of Reaction Conditions for the Oxidation of Benzyl Alcohola

entry	catalyst	oxidant	conditions	time (h)	yield (%)
1	RCAC	airb	PhMe/120 ° C/LED	12	35
2	RCAC	airb	PhMe/120 ° C/dark	12	10
3	Cu0.9Fe0.1@RCAC	airb	PhMe/120 ° C/LED	2	98
4	Cu0.5Fe0.5@RCAC	airb	PhMe/120 ° C/LED	2	62
5	Cu0.1Fe0.9@RCAC	airb	PhMe/120 ° C/LED	2	32
6	Cu@RCAC	airb	PhMe/120 ° C/LED	2	25
7	Fe@RCAC	airb	PhMe/120 ° C/LED	2	20
8	Cu0.9Fe0.1@RCAC	H2O2c	PhMe/120 ° C/LED	2	96
9	Cu0.9Fe0.1@RCAC	TBHPd	PhMe/120 ° C/LED	2	99
10	Cu0.9Fe0.1@RCAC	airb	H2O/120 ° C/LED	2	
11	Cu0.9Fe0.1@RCAC	airb	EtOH/120 ° C/LED	2	38
12	Cu0.9Fe0.1@RCAC	airb	THF/120 ° C/LED	2	41
13	Cu0.9Fe0.1@RCAC	airb	dioxane/120 ° C/LED	2	58
14	Cu0.9Fe0.1@RCAC	airb	DMSO/120 ° C/LED	2	67
15	Cu0.9Fe0.1@RCAC	airb	PhMe/120 ° C/dark	2	52

脚注（逐字）：aConditions: benzyl alcohol (0.5 mmol), catalyst (50 mg), total metal loading adjusted to 10 wt %, solvent (2 mL), 10 W LED bulb.bAir flow using air pump.cH2O2 (1.5 mmol).dTBHP (1.5 mmol).

正文对应句（逐字）：Next, the catalytic activity of the CuxFe1 – x@RCAC materials was investigated in the oxidation of primary and secondary alcohol. Initially, we have examined the oxidation of benzyl alcohol as a model reaction under light-emitting diode (LED) light. When a mixture of benzyl alcohol (0.5 mmol) and pris

本卡为 NCBI PMC 开放获取 XML
的逐字摘录（标题、作者、表格行列、脚注、正文句均未改写），用于参照集候选与覆盖面登记；未逐条人工核验，标记
imported_unchecked，不写入判定记录、不参与 ChemCritic 引擎的可比性判定。对外引用一律回到 DOI 原文核对页码与表号。